

**Factors Associated with Self-Reported Prostate-Specific Antigen (PSA) Testing and
Changes in Its Prevalence Among U.S. Males Aged 40 Years and Older: A Comparative
Analysis of 2014 and 2024 BRFSS Data**

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INTRODUCTION

Prostate-specific antigen (PSA) testing remains a widely used approach for early detection of prostate cancer and other prostate-related abnormalities, yet its use has been shaped by evolving testing recommendations and longstanding debate regarding its benefits and limitations.

Recommendations from the U.S. Preventive Services Task Force (USPSTF), particularly the 2012 and 2018 guideline changes, have influenced provider practices and patient decision-making regarding PSA testing, contributing to shifts in testing prevalence over time (Leapman et al., 2022). Despite its continued role in preventive care, PSA testing utilization varies substantially across population groups, raising important public health questions regarding equitable access to testing and informed decision-making.

Existing research suggests that sociodemographic and structural factors, including race/ethnicity, education, and health insurance coverage, influence PSA testing utilization and contribute to disparities in testing practices (Washington et al., 2022). Prior studies have documented declines in PSA testing following changing recommendations and persistent disparities across populations (Leapman et al., 2022; Lillard et al., 2022). However, much of this literature has relied on single time-point analyses and less is known about how PSA testing prevalence and associated disparities may have changed across recent guideline eras using nationally representative data. Understanding these changes is important for informing public health approaches to equitable and evidence-based testing practices. Using pooled Behavioral Risk Factor Surveillance System (BRFSS) data from 2014 and 2024, this study examines changes in self-reported PSA testing prevalence and associations with key sociodemographic factors among U.S. males aged 40 years and older. The purpose of this research is to generate evidence that may inform PSA testing recommendations, improve understanding of PSA testing disparities, and support equitable prostate cancer prevention strategies.

Background

Prostate cancer remains an important public health concern among men, particularly because risk increases with age and outcomes differ across racial and socioeconomic groups (Chowdhury-Paulino et al., 2022). Although this study focuses on PSA testing rather than prostate cancer outcomes, prostate cancer trends provide important context for understanding why testing practices remain clinically and publicly relevant. Recent national cancer statistics indicate prostate cancer incidence has increased approximately 3% annually in recent years, with rising

diagnoses of advanced-stage disease raising renewed concern regarding early detection strategies (Siegel et al., 2026). In addition, Black men continue to experience disproportionately higher prostate cancer incidence and mortality compared with other racial groups, underscoring persistent inequities relevant to testing and prevention efforts (Chowdhury-Paulino et al., 2022; Lillard et al., 2022).

Prostate-specific antigen (PSA) testing is a blood test used to identify elevated PSA levels that may indicate prostate cancer or other prostate-related conditions. While PSA testing may facilitate earlier detection, interpretation of elevated PSA values is complex because benign conditions may also contribute to abnormal results (Perdana et al., 2016; Rawla, 2019). Concerns regarding false positives, overdiagnosis, and overtreatment have contributed to longstanding debate about routine testing and have shaped evolving clinical recommendations (Eastham, 2017). These controversies have increasingly shifted emphasis toward individualized risk assessment and shared decision-making in testing practice (Basin et al., 2023).

USPSTF recommendations have changed substantially over time in response to these debates. The 2012 Grade D recommendation discouraged routine PSA-based testing, whereas the 2018 Grade C recommendation emphasized individualized decision-making for men aged 55-69 years (Leapman et al., 2022). These guideline shifts have been linked to changes in testing behaviors; national data suggest PSA testing declined substantially following the 2012 recommendation, including an estimated 46% decline in past-year PSA testing between 2008 and 2020, followed by only partial rebound thereafter (Merrill & Gibbons, 2024). The selection of 2014 and 2024 therefore provides two comparison points across major USPSTF guideline eras and allows examination of changes in PSA testing prevalence and associated sociodemographic patterns across two distinct post-guideline periods.

Emerging evidence suggests PSA testing behaviors are shaped not only by clinical recommendations, but also by broader sociodemographic and structural influences, including healthcare access, insurance coverage, and shared decision-making processes (Burnett et al., 2023; Washington et al., 2022). Recent BRFSS-based research has also documented declines in testing utilization over time, including reductions from 51.2% to 36.4% in biennial PSA testing prevalence among men aged 40 years and older in one population-based analysis between 2012 and 2020 (Fullwood et al., 2023). Together, these findings suggest PSA testing utilization may

be shaped by both individual and structural factors and provide an important foundation for examining changes in self-reported PSA testing prevalence and associated disparities over time.

Statement of the Problem

Prostate cancer is the second most frequently diagnosed cancer among men and the fifth leading cause of death worldwide (Rawla, 2019). It is treatable in early stages to help prevent progression. However, delayed diagnosis is associated with more advanced stages which are incurable and increasingly challenging to treat, resulting in increased mortality rates. Prostate cancer testing is crucial for early detection of prostate cancer and other prostate-related abnormalities. PSA testing may facilitate early detection, which can improve opportunities for timely treatment and may contribute to improved outcomes. Key risk factors influencing testing behaviors include physician recommendation, age, family history, socioeconomic status, health access, and knowledge or awareness. While timely diagnosis and testing are critical for improving long-term prostate cancer outcomes, many men lack appropriate knowledge on the risks and importance of PSA testing. Additionally, evolving USPSTF testing recommendations may contribute to variation in clinical practice and physician recommendations.

Ideal

In an ideal world, PSA testing would be guided by evidence-based clinical recommendations, with equitable access to preventive health services for men aged 40 years and older regardless of socioeconomic status or access to healthcare. Patients would engage in informed, shared decision-making with healthcare providers, allowing for individualized testing decisions based on risk factors, preferences, and clinical guidelines. Under these conditions, PSA testing practices would be consistently applied, disparities in testing utilization across sociodemographic groups would be minimized, early detection opportunities would improve, prostate cancer disparities would narrow, and health outcomes would improve across populations.

Reality

Unfortunately, the reality is far from ideal. PSA testing prevalence has declined by approximately 9.5% since 2012, with the largest decline seen among Black men at 11.6% compared to a 9.3% decline in White men (Lillard et al., 2022). Testing practices vary considerably across populations due to differences in sociodemographic characteristics and access to healthcare. In the U.S., Black men have PSA testing rates of approximately 30% which is lower than the 33% PSA testing rates seen in White men (Lillard et al., 2022). Individual risk

factors including race, education level, and insurance status influence an individual's likelihood of receiving PSA testing, reflecting broader social determinants of health (Burnett et al., 2023; Washington et al., 2022). Moreover, inconsistencies in provider recommendations, variability in patient-provider communication, and differences in awareness of testing guidelines contribute to unequal uptake of PSA testing (Leapman et al., 2022).

Consequences

The consequences of continued reduced uptake of PSA testing are detrimental. Ongoing unequal access to testing and differences in utilization may contribute to continued delayed diagnosis and poorer prostate cancer outcomes among U.S. men, disproportionately affecting more vulnerable populations. The human toll includes preventable deaths, reduced quality of life, and lost economic productivity. Healthcare systems will face the burden of care costs and limited resources associated with poorer long-term prostate cancer outcomes and continued management of advanced-stage disease. Public health efforts must therefore focus on generating evidence that can inform targeted testing interventions and evidence-based PSA testing recommendations—identifying key risk factors influencing inequities in PSA testing, as well as understanding their complex interconnection with current PSA testing practices and changes in PSA testing outcomes. Understanding these nuanced relationships is essential for designing effective policies, programs, and guidelines that promote prostate cancer testing equity.

Significance of the Study

This study addresses a critical gap in the literature by examining changes in PSA testing prevalence and associated factors across two distinct time points spanning major USPSTF guideline changes using nationally representative data from the Behavioral Risk Factor Surveillance System (BRFSS). Previous research has established the risk factors for prostate cancer and has documented PSA testing practices, prevalence, controversies, and outcomes. However, much of this evidence derives from studies with limited population diversity or from older data that may not capture contemporary testing practices, state-level variation, or changes following evolving testing recommendations. Furthermore, few studies have examined how major PSA testing guideline changes may have influenced testing behaviors over time. Moreover, less is known about how these disparities may have changed across recent guideline eras. By analyzing BRFSS data collected across all 50 states, the District of Columbia, and three U.S. territories at two distinct time points, this research will provide current, population-based

estimates of associations between multiple sociodemographic factors (e.g., insurance status, race/ethnicity, and education level) and PSA testing while accounting for important confounders. The study's focus on disparities across racial, ethnic, and socioeconomic subgroups is particularly timely. Recent evidence demonstrates that prostate cancer testing practices and outcomes vary substantially across communities and may contribute to delayed diagnosis at the population level (Lillard et al., 2022). Additionally, access to quality and affordable care influences shared decision-making processes, contributing to variation in PSA testing prevalence (Basin et al., 2023). Understanding how these structural factors intersect with individual-level risk factors to influence prostate cancer testing can inform the development of public health interventions to address key social and structural determinants influencing testing disparities. Furthermore, this research will contribute to public health practice by generating actionable evidence for policymakers, healthcare providers, and community organizations. By identifying population subgroups with lower testing uptake and factors associated with PSA testing utilization, findings may help guide resource allocation, targeted interventions, and strategies to promote informed decision-making around PSA testing. The use of BRFSS data ensures that results will be immediately relevant to state and local public health agencies already familiar with this surveillance system, facilitating translation of findings into practice and informing future testing policy and research. Findings may also support broader public health efforts to address testing inequities driven by social and structural determinants of health, promote equitable access to preventive care and early detection, and potentially improve health outcomes while reducing long-term healthcare costs.

What is Already Known on this Topic

- PSA testing may facilitate earlier detection of prostate cancer and other prostate-related abnormalities.
- PSA testing recommendations have changed substantially over time, influencing testing practices.
- PSA testing frequencies have generally declined following the 2012 USPSTF recommendation.
- PSA testing utilization differs substantially across racial, ethnic, and socioeconomic groups, with documented disparities among disadvantaged populations.

- Sociodemographic and structural factors contribute to disparities in PSA testing utilization.

What this Study is Expected to Add

- Compares factors associated with PSA testing utilization and changes in prevalence across two distinct time points (2014 and 2024).
- Provides updated national estimates of changes in PSA testing prevalence using 2014 and 2024 BRFSS data.
- Examines how testing patterns and associated factors may have changed following major guideline shifts.
- Generates evidence to inform equitable testing interventions, public health policy, and strategies to improve outcomes while reducing long-term healthcare burden

LITERATURE REVIEW

Prostate Cancer Testing Guidelines and Testing Trends

Prior research suggests PSA testing patterns have changed substantially in response to evolving testing recommendations. The 2012 USPSTF Grade D recommendation was associated with declines in PSA testing utilization, whereas the 2018 Grade C revision re-emphasized individualized testing decisions among men aged 55-69 years and may have influenced subsequent shifts in testing practices (Leapman et al., 2022). However, findings vary regarding the extent to which testing behaviors rebounded following these guideline changes.

Evidence also suggests changes in PSA testing have not occurred uniformly across populations. Some studies document sustained declines in testing utilization following guideline shifts, while others suggest more modest changes influenced by provider practices, patient preferences, and healthcare access (Fullwood et al., 2023). Together, this literature indicates PSA testing trends reflect both evolving recommendations and broader factors influencing preventive healthcare utilization.

Sociodemographic and Structural Factors Associated with PSA Testing

A substantial body of literature suggests PSA testing utilization is associated with both sociodemographic and structural determinants. Race/ethnicity, educational attainment, insurance coverage, and healthcare access have all been linked to variation in uptake, although findings

regarding the magnitude of these associations differ across studies (Burnett et al., 2023; Washington et al., 2022). Evidence generally suggests individuals with greater access to care and higher socioeconomic resources may be more likely to engage in preventive testing, while structural barriers may limit utilization among underserved populations.

Although many studies report positive associations between healthcare access and PSA testing utilizations, findings are less consistent regarding the independent contribution of educational attainment after accounting for other sociodemographic factors. Shared decision-making has also emerged as an important influence on PSA testing behaviors. Recent studies suggest access to informed provider-patient discussions may vary across sociodemographic groups, potentially contributing to disparities in testing uptake (Basin et al., 2023). Collectively, this literature suggests PSA testing behaviors are shaped not only by individual risk factors but also by broader structural influences relevant to public health equity.

Disparities in PSA Testing Utilization and Testing Outcomes

Disparities in PSA testing utilization remain a consistent theme across the literature. Studies have documented differences in testing uptake across racial, ethnic, socioeconomic groups, with inequities in access, utilization, and shared decision-making often reflecting broader social determinants of health (Chowdhury-Paulino et al., 2022). Although these disparities are well documented, evidence is mixed regarding how they may have shifted following recent guideline changes.

Variation in implementation of testing recommendations across providers and healthcare settings may further shape how disparities in PSA testing are experienced across populations.

Understanding how disparities in utilization intersect with changing guidelines and structural barriers remains important for interpreting trends in PSA testing and identifying areas for more equitable testing approaches.

Gaps in Existing Literature

While prior research has examined prostate cancer epidemiology, PSA testing practices, and sociodemographic correlates of testing utilization, several important gaps remain. Much of the literature has relied on single time-point cross-sectional analyses, limiting understanding of how PSA testing prevalence and associated factors may have changed over time, particularly across major USPSTF guideline eras (Burnett et al., 2023; Lillard et al., 2022). In addition, relatively few studies have conducted repeated cross-sectional analysis using nationally representative data

to compare changes in self-reported PSA testing across distinct time points following major guideline changes.

Although disparities in PSA testing utilization by race/ethnicity, education, and healthcare access are well documented, less is known about whether these disparities have changed over time or whether associations between sociodemographic factors and PSA testing vary across survey years (Leapman et al., 2022; Washington et al., 2022). These gaps limit understanding of potential shifts in testing patterns and disparities over time.

Finally, relatively few studies have used recent nationally representative BRFSS data to examine factors associated with self-reported PSA testing across repeated cross-sectional samples. This study seeks to address these gaps by evaluating changes in PSA testing prevalence and associated factors using pooled 2014 and 2024 BRFSS data.

Purpose and Objective

Purpose

The purpose of this study is to generate evidence that may inform evidence-based PSA testing recommendations and public health strategies aimed at improving equitable and informed PSA testing utilization among U.S. males aged 40 years and older. By examining changes in PSA testing prevalence and associated factors across major USPSTF guideline eras, this study seeks to support efforts to reduce testing disparities and improve prostate cancer prevention outcomes.

Objective(s)

- To compare the prevalence of self-reported PSA testing among U.S. males aged 40 years and older in 2014 and 2024.
- To assess associations between sociodemographic factors, including health insurance status, education level, and race/ethnicity, and self-reported PSA testing among U.S. males aged 40 years and older.
- To examine whether associations between sociodemographic factors and self-reported PSA testing vary across survey years, indicating potential changes in testing disparities over time.

Research Question(s)

Is there an association between sociodemographic factors and self-reported PSA testing among U.S. males aged 40 years and older, and do PSA testing prevalence and these associations differ between 2014 and 2024?

Hypotheses

Null Hypothesis (H_0)

There is no significant association between sociodemographic factors, including health insurance status, education level, and race/ethnicity, and self-reported PSA testing among U.S. males aged 40 years and older, and there is no significant difference in PSA testing prevalence between 2014 and 2024.

Alternative Hypothesis (H_1)

Sociodemographic factors, including health insurance status, education level, and race/ethnicity, are significantly associated with self-reported PSA testing among U.S. males aged 40 years and older, and PSA testing prevalence differs significantly between 2014 and 2024.

Secondary Hypothesis

The associations between sociodemographic factors and self-reported PSA testing differ by survey year among U.S. males aged 40 years and older, indicating changes in disparities between 2014 and 2024.

METHODS

Data Source and Collection

This study will utilize data from the 2014 and 2024 Behavioral Risk Factor Surveillance System (BRFSS), a nationally representative, state-based health-related telephone survey conducted by the Centers for Disease Control and Prevention (CDC) in collaboration with state and territorial health departments. The Behavioral Risk Factor Surveillance System is the nation's premier system of health-related telephone surveys that collects state data on health-related risk behaviors, chronic health conditions, health care access, and use of preventive services among noninstitutionalized U.S. adults aged 18 years and older (Centers for Disease Control and Prevention, 2024a). BRFSS data is widely used to examine preventive care utilization, including prostate-specific antigen (PSA) testing trends (Fullwood et al., 2023).

Survey Design and Methodology

The Behavioral Risk Factor Surveillance System (BRFSS) employs a complex, multistage sampling design that incorporates stratification and weighting to produce representative

estimates at both state and national levels. Sampling weights are applied to account for differences in selection probability and to adjust for the distribution of sociodemographic characteristics among respondents (Merrill & Gibbons, 2024). The survey utilizes random-digit-dialing (RDD) as its primary data collection method (Fullwood et al., 2023). To improve coverage, validity, and overall data quality, BRFSS uses a dual-frame approach that includes both landline and cellular telephone samples (Centers for Disease Control and Prevention, 2023). This approach enhances representativeness and reduces coverage bias observed in earlier survey years. The BRFSS questionnaire consists of three components: (1) core questions assessing demographic characteristics, health conditions, and health behaviors administered across all states and territories; (2) optional modules on specific topics, such as prostate cancer testing; and (3) state-added questions developed to address local public health priorities (Merrill & Gibbons, 2024). In this analysis, BRFSS questions related to prostate-specific antigen (PSA) testing will be used to define the primary outcome, while selected demographic and healthcare access variables will be included to examine factors associated with PSA testing among adult males in the United States.

Weighting Methodology

In 2011, BRFSS adopted a revised weighting methodology known as iterative proportional fitting, or raking, which replaced the previous poststratification approach. This update enabled the incorporation of cellular telephone surveys and allowed for the inclusion of additional demographic characteristics in the weighting process (Centers for Disease Control and Prevention, 2023). Weighting is used to ensure that the sample is representative of the overall population by accounting for selection probabilities, adjusting for sociodemographic distributions, and correcting for nonresponse. Differences between landline and cellular samples are also incorporated into the weighting process to address variation in coverage across telephone modes.

Weighting variables include age, race, sex, education, ethnicity, marital status, home ownership, sub-state region, and telephone ownership (landline only, cell phone only, dual user) (Iachan et al., 2016). These weighting variables are used in the raking procedure to generate the final survey weight (_LLCPWT), which is assigned to each respondent (Centers for Disease Control and Prevention, 2023). Application of the final weight is essential for producing valid population estimates and appropriate statistical inferences when analyzing BRFSS data.

Researchers analyzing large national health surveys such as BRFSS, the National Health Interview Survey (NHIS), the National Health and Nutrition Examination Survey (NHANES), and the National Survey on Drug Use and Health (NSDUH) routinely apply weighting procedures to generate nationally representative estimates of health-related outcomes (Iachan et al., 2016). This study will follow this methodological approach by applying survey weights and accounting for the complex sampling design through appropriate survey procedures in RStudio version 2026.01.0+392.

Survey Administration and Quality Control

According to BRFSS guidelines, Computer-Assisted Telephone Interview (CATI) systems are used to conduct interviews. CATI automates the administration of core and module questions and provides standardized scripts for both core and state-added questions (Centers for Disease Control and Prevention, 2023). This system enhances consistency across interviews and supports data quality. In addition, BRFSS provides interviewer training for state health personnel and contractors, along with ongoing performance evaluations through randomized interview evaluations. These procedures collectively strengthen validity and reliability of the collected data.

Study Population

This study will utilize publicly available data from the BRFSS for survey years 2014 and 2024. The target population for this study is non-institutionalized adult males aged 40 years and older residing in the United States. Restricting the sample to males aged 40 years and older ensures that the analysis focuses on a population more relevant to PSA testing practices and guideline-based testing discussions. The inclusion of multiple survey years allows for the evaluation of PSA testing patterns across a repeated cross-sectional comparative study design, where each survey year represents an independent snapshot of the population. This temporal scope enables a comparative assessment of PSA testing behaviors and their associations with sociodemographic and healthcare-related factors across 2014 and 2024, while maintaining nationally representative estimates for each survey year. BRFSS collects data across all 50 states, the District of Columbia, and U.S. territories, with over 400,000 adults interviewed annually (Centers for Disease Control and Prevention, 2024b). The sample size for each year is large enough to provide adequate statistical power to assess the prevalence of PSA testing.

Inclusion Criteria

Male respondents aged 40 years and older who participated in BRFSS survey years 2014 and 2024 and provided valid responses to questions measuring the independent variables, dependent variables, and confounders will be eligible for inclusion in this study.

Exclusion Criteria

Female respondents will be excluded from this study, as prostate cancer testing is specific to testing males. Respondents with missing, incomplete, or invalid data for the primary outcome variable (PSA testing), key independent variables, or covariates included in the analysis will also be excluded to ensure data completeness and validity of the results.

Variables and Measurement

Independent Variable: Insurance status

Insurance status will be measured using BRFSS core question: “What is the current primary source of your health care coverage?” This question has multiple categorical response options capturing different types of insurance coverage. For this study, insurance status will be recoded into a binary variable in RStudio to assess its association with PSA testing. Responses will be coded as 0 for uninsured and 1 for insured individuals. Additionally, responses with missing responses, refusals, and “don’t know/not sure” will be excluded from analysis.

Independent Variable: Education level

Education level will be measured using BRFSS core question: “What is the highest grade or year of school you completed?” This question includes multiple categorical response options representing increasing levels of educational attainment. For this study, education level will be analyzed as a categorical variable and recoded as a factor variable in RStudio. One category will be selected as the reference group, and each remaining category will be compared to the reference group in a logistic regression model to estimate adjusted associations with PSA testing across levels of educational attainment.

Independent Variable: Race/ethnicity

Race will be measured using BRFSS core question: "Which one or more of the following would you say is your race?" This variable includes multiple categorical response options representing major racial groups. For analysis, race will be recoded as a factor variable in RStudio. Categories will be grouped into meaningful classifications, with one category selected as the referent group. Each category will be compared to the reference group in a logistic regression model to estimate adjusted associations with PSA testing across racial groups.

Dependent Variable: Self-reported PSA testing

Prostate-specific antigen testing will be measured using the BRFSS module question: “Have you ever had a P.S.A. test?” Responses are self-reported and reflect whether an individual has undergone PSA testing at any point in their lifetime. For this analysis, PSA testing will be recoded as a binary variable in RStudio. Respondents who report having had a PSA test will be coded as 1 (yes) and those with no previous PSA test will be coded as 0 (no). This binary outcome will be used in a logistic regression model to assess associations between PSA testing and the selected independent variables. Responses with missing responses, refusals, and “don’t know/not sure” values will be excluded from analysis.

Covariates

Age will be used in a logistic regression model to control for its potential confounding effect on the association between the independent variables and PSA testing. In BRFSS 2024, age is collected as a continuous variable based on respondents’ self-reported age in years in response to the core question: “What is your age?” For this analysis, age will be grouped into categorical age ranges and analyzed as a factor variable in RStudio. This approach allows for adjustment across age groups while accounting for potential differences in PSA testing behavior by age.

Statistical Analysis

All statistical analyses will be conducted using RStudio version 2026.01.0+392, with appropriate survey procedures that account for the complex sampling design of the BRFSS. Data from 2014 and 2024 BRFSS will be appended into a pooled analytic dataset for analysis. Only survey items measured consistently across both study years will be included to ensure comparability across time points. Weighting will be applied in all analyses to produce population-representative estimates. Statistical significance will be set at $\alpha \leq 0.05$ for all hypothesis tests, with 95% confidence intervals reported for all estimates. Odds ratios will also be calculated and reported for logistic regression models.

Descriptive Analysis

Descriptive statistics will be used to summarize the study population for each survey year (2014 and 2024) using frequency tables. Weighted frequencies and weighted percentages will be calculated for categorical variables, including PSA testing status, insurance status, education level, and race. All reported estimates will incorporate BRFSS survey weights to ensure nationally representative results. Comparisons of PSA testing prevalence and sociodemographic

characteristics across categories of the independent variables will be conducted for each year by examining the differences in distributions. These analyses will provide an overview of the sample characteristics and allow comparison with the broader U.S. adult populations for both survey years.

Bivariate Analysis

Bivariate analyses will be conducted separately for each survey year (2014 and 2024) to examine the unadjusted associations between PSA testing and each independent variable (insurance status, education level, and race), as well as the covariate age. Chi-square tests will be used to assess these associations between PSA testing and each categorical variable. Weighted prevalence estimates and corresponding 95% confidence intervals will be calculated for PSA testing across each independent variable and within subgrouping comparisons. These analyses will identify crude associations and describe patterns in PSA testing across sociodemographic and healthcare-related characteristics within each survey year. Statistical significance will be evaluated at an alpha level of 0.05, and results will be reported as weighted proportions with corresponding p-values to facilitate comparison between 2014 and 2024. In addition, chi-square tests will be conducted to assess differences in PSA testing prevalence by survey year.

Multivariable Analysis

A pooled multivariable logistic regression model will be conducted using combined BRFSS data from 2014 and 2024 to examine adjusted associations between PSA testing and the independent variables (insurance status, education level, and race), while controlling for age. Survey year will be included as an indicator variable to assess differences in PSA testing prevalence between 2014 and 2024. Survey weights will be applied to account for the complex sampling design of BRFSS, ensuring nationally representative estimates. Adjusted odds ratios (AORs) with corresponding 95% confidence intervals will be reported for all associations.

The logistic regression model will be specified as follows:

$$\begin{aligned} \text{logit}(P(\text{PSA testing})) \\ = \beta_0 + \beta_1(\text{year}) + \beta_2(\text{insurance status}) + \beta_3(\text{education level}) + \beta_4(\text{race}) \\ + \beta_5(\text{age}) \end{aligned}$$

This model will assess both sociodemographic factors associated with PSA testing and differences in testing prevalence between survey years.

Interaction Analysis

To examine whether associations between sociodemographic factors and PSA testing differ by survey year, interaction terms between survey year and selected independent variables (e.g., insurance status, education level, and race/ethnicity) will be added to the pooled multivariable logistic regression model. Significant interaction terms will be interpreted as evidence that disparities in PSA testing may have changed between 2014 and 2024.

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